

Gizem Toplu Tutay

Last updated: October 1, 2025

☎ +1 737 262 3389 • ✉ gizem@utexas.edu • 🌐 gizemtt.github.io

EDUCATION

The University of Texas at Austin

Austin, TX

PhD in Operations Research and Industrial Engineering

2021 – Spring 2026 (expected)

Thesis: Stochastic and Data-Driven Optimization for Equity-Focused Resilience in Power and Healthcare Systems (Advisors: Prof. John J. Hasenbein and Prof. Erhan Kutanoglu)

The University of Texas at Austin

Austin, TX

MS in Operations Research and Industrial Engineering

2024

Stanford University and DeepLearning.AI

Online

Machine Learning Specialization

Certification, 2023

Bogazici University

Istanbul, Turkey

BS in Civil Engineering (Highest Honors, ranked 3rd)

2008 – 2013

Thesis: Optimizing Wind Energy and Storage: A Financial and Operational Investment Analysis

RESEARCH INTERESTS

Methodology: Integer programming, network optimization, multi-stage stochastic optimization, robust optimization, machine learning

Applications:

- Modeling and optimization of critical systems to enhance resilience under extreme events
- Data analytics and stochastic modeling of disaster impacts, social vulnerability, and risk propagation across interconnected systems
- Policy-oriented decision support for sustainable and socially responsible investment and resource allocation under uncertainty

PUBLICATIONS

Published Journal Papers:

- Toplu-Tutay, G., Hasenbein, J. J., Kutanoglu, E., “Impact of Power Outages Depends on Who Loses It: Equity-Informed Grid Resilience Planning via Stochastic Optimization,” *Socio-Economic Planning Sciences*, Vol. 95, 102036, pp. 1–16, October 2024
- Kim, K. Y., Toplu-Tutay, G., Kutanoglu, E., Hasenbein, J. J., “A Stochastic Optimization Model for Patient Evacuation from Health Care Facilities during Hurricanes,” *International Journal of Disaster Risk Reduction*, Vol. 108, 104518, pp. 1–18, June 2024

Peer Reviewed Conference Papers:

- Das, J., Toplu-Tutay, G., Gulen, G., and Young, M., “Power Grid Dispatch Modeling for the Prospective Current Trends Scenario for the ERCOT Service Area of Texas,” *IEEE Texas Power and Energy Conference (TPEC)*, pp. 1–6, 2025
- Toplu-Tutay, G., Hasenbein, J. J., Kutanoglu, E., “Scenario-based optimization model for long-term healthcare infrastructure resilience against flooding,” *IISE Annual Conference.Proceedings*,

pp. 1–6, 2022

Work in Preparation:

- Multi-Objective Healthcare System Resilience: Mitigate Now or Respond Later? (with John J. Hasenbein and Erhan Kutanoglu; and Matt Kammer-Kerwick, Bureau of Business Research / IC² Institute)
- Quantifying the Human Cost of Storms: Social Vulnerability as a Driver of Disaster Mortality
- Combating Illicit Supply Chains through Analytics for Social Good (with Matt Kammer-Kerwick, Bureau of Business Research/IC² Institute)
- Resilient and Sustainable Energy Operations: Multi-Objective Planning under Policy and Decarbonization Scenarios (with Jani Das and Michael H. Young, Bureau of Economic Geology)

RESEARCH AND INDUSTRY EXPERIENCE

The University of Texas at Austin

Doctoral Research Assistant

Austin, TX

Jan 2021 – present

Bureau of Business Research/IC² Institute - funded research (2025–2026)

- Leading a perspective paper, “Combating Illicit Supply Chains through Analytics for Social Good,” synthesizing interdisciplinary research to define future directions at the intersection of operations research, policy, and social justice

University Graduate Continuing Fellowship Research (2024–2025)

- Developing a data-driven framework to quantify long-term storm-induced mortality by integrating historical storm data with county-level death records
- Analyzing geographic, urbanization, and demographic patterns to identify factors shaping mortality risk across U.S. counties

Bureau of Economic Geology - Comparing Electricity Options (CEO) project (2023-2024)

- Developed a mixed-integer programming (MIP) model for capacity expansion planning to optimize the future energy mix, ensuring grid reliability under energy transition scenarios with the gradual integration of wind, solar, and battery storage technologies

Bureau of Business Research/IC² Institute - funded research (2022-2023)

- Developed a two-stage stochastic integer programming model for disaster mitigation investments in transmission grids, with a focus on substation hardening
- Created equity metrics to incorporate the uneven impacts of prolonged power outages on socially vulnerable communities
- Formulated equity-informed models and a justice model inspired by the U.S. government’s Justice40 Initiative
- Conducted a case study on the Texas grid’s resilience during hurricane-induced flooding

National Science Foundation (NSF)-funded research (2021-2022)

- Developed a two-stage stochastic integer programming model for staging area location(s) and resource allocation, considering the uncertainty in patient evacuation needs due to the uncertain landfall and precipitation of an emerging hurricane
- Integrated the decision-making model with physics-based hydrological models for inland flooding and storm surge
- Conducted a case study on Hurricane Harvey’s impact on the Texas coastline

National Renewable Energy Laboratory (NREL)

Ph.D. Research Intern

Golden, CO

May 2024 – August 2024

- Contributed to a project optimizing electric vehicle managed charging (EVMC) to improve bulk power grid reliability, affordability, and decarbonization
- Assisted in integrating EVMC into ReEDS, NREL's flagship capacity planning model (MIP-based), and validated it using load profiles and charging flexibility data from the TEMPO model
- Developed scripts for translating TEMPO data to ReEDS format and generating interpretable outputs from optimal solutions and constraint marginals
- Collaborated with a large, distributed team while adhering to coding standards, leveraging version control tools like Git and GitHub for code development and management
- Utilized High-Performance Computing (HPC), working remotely with cloud-based infrastructure

GISEMAI

Ankara, Turkey

Founder

Jan 2018 – Dec 2020

- Led an e-commerce startup, managing global logistics, supply chain management, and operations
- Applied inventory management techniques, including *just-in-time* and the *newsvendor* model, to ensure effective stock control and demand forecasting
- Managed end-to-end processes, from manufacturing to warehousing, sales, and delivery.

CLK Energy Investment

Istanbul, Turkey

Senior Strategic Planning Analyst

Jun 2015 – Nov 2017

- Led a strategic planning team, collaborating with departments across the enterprise (trading, finance, onsite technical teams)
- Developed and applied time series forecasting models to predict electricity demand, enabling data-driven strategic decisions
- Led cross-functional initiatives to shape or inform the company's long-term plans and decisions

ENERJISA

Istanbul, Turkey

Key Account Manager

Sep 2013 – Mar 2014

- Acquired and managed key accounts (high-electricity demand companies and industrial facilities) for Turkey's largest electricity generation and distribution company
- Analyzed energy markets, negotiated high-value contracts, and collaborated with onsite technical teams to support key accounts in achieving optimal energy solutions

HONORS AND AWARDS

- \$52K 1 year University Graduate Continuing Fellowship for outstanding academic accomplishments (2024-2025)
- IC² Institute Seed Grant, *Towards a Well-Being Economy: Advancing Equity in Social, Economic, and Health Systems* (2022-2023)
- NSF Award #1940308, *CoPe EAGER: Addressing Human-Centric Decision-Making Challenges from Coastal Hazards via Integrated Geosciences Modeling and Stochastic Optimization* (2021-2022) (Participated under advisors)
- Graduated *summa cum laude* (with highest honors), B.S. (2013)
- Academic excellence scholarship for outstanding performance (2008-2013)

TEACHING EXPERIENCE

Department of Mechanical Engineering - Cockrell School of Engineering

Teaching Assistant for Engineering Statistics

Fall 2022

- Upper-division course on probability, distribution theory, data analysis, statistics, interval estimation, hypothesis testing, and statistical quality control.
- Designed and led recitation classes, held weekly office hours, graded assignments and exams.
- Responsible for 25 students

Department of IROM - McCombs School of Business

Teaching Assistant for Learning Structures & Time Series

Spring 2021

- Graduate-level course on unsupervised machine learning (principal component analysis, cluster analysis, factor analysis) and time series forecasting (ARIMA, autoregression).
- Held weekly office hours and graded assignments and exams.
- Responsible for 105 students

Talks

- Presented *Equity-Informed Power Grid Resilience Planning via Stochastic Optimization* at:
 - INFORMS Annual Meeting, Seattle, October 2024 (Poster)
 - Society for Risk Analysis Annual Meeting, Austin, December 2024
 - Operations Research and Industrial Engineering Seminar, The University of Texas at Austin, April 2025
- Presented *Disparate Impacts of Extreme Flooding And Power Grid Failures in Different Communities*, INFORMS Annual Meeting, Indianapolis, October 2022
- Presented *A Scenario-based Optimization Model for Long-term Healthcare Infrastructure Resilience against Flooding*, IISE Annual Conference, Seattle, May 2022
- Led annual seminars at CLK Energy Investment. Presented on electricity market developments to an audience managing 25% of Turkey's total electricity demand (2015, 2016, 2017).

PROFESSIONAL SERVICE

- Reviewer for *Applied Energy* and *IEEE Transactions on Energy Markets, Policy, and Regulation* (2023–present)
- Member of INFORMS and the International Association for Energy Economics (IAEE)
- Chair, Tenant Advisory Board for University Housing and Dining, The University of Texas at Austin (2023–present)
- Organizer and committee member for multiple IAEE international conferences and symposia, including the Eurasian Conferences and the Southeast Europe Symposium (2016–2019)
- Organizer of IAEE Summer Schools on Electricity Markets and Regulation, Istanbul, Turkey (2015–2016)

TECHNICAL SKILLS

Programming: Python, R, MATLAB, SQL

Optimization Tools: GAMS, Pyomo, Gurobi, CPLEX

Methodologies: Mixed-integer programming, network optimization, multi-stage stochastic and robust optimization

Machine Learning: Supervised and unsupervised learning, deep learning, reinforcement learning